DAY 34 LAB: Standard ACLs

Goals:

1.) Configure OSPF on R1 and R2 to allow full connectivity between the PCs and servers. **[DONE]**

2.) Configure standard numbered ACLs on R1 **[DONE]** and standard named ACLs on R2 to fulfill **[DONE]**
the following network policies:
→ Only PC1 and PC3 can access 192.168.1.0/24 **[DONE]**
→ Hosts in 172.16.2.0/24 can't access 192.168.2.0/24 **[DONE]**
→ 172.16.1.0/24 can't access 172.16.2.0/24 **[DONE]**
→ 172.16.2.0/24 can't access 172.16.1.0/24 **[DONE]**

Important commands that are used:

Full form command	Purpose
router ospf 1	To enter router configuration for OSPF and its process ID
passive-interface g0/0	Used for end hosts intermediary devices to disable receiving “OSPF Hello messages (useful to find potential router neighbors)” unnecessary traffic for end hosts.
network 192.168.1.0 0.0.0.255 area 0	To enable OSPF in this network and specify its area
ip access-list standard 1	To create the numbered standard access-list
Ip access-list standard FOR-192.168.1.0/24	To create the named standard access-list
1 {deny | permit} 192.168.1.0 0.0.0.255	To create ACE (Access Control Entry). In this example you specified the sequence number, permit/deny, network and its wildcard mask.
{deny | permit} 192.168.1.0 0.0.0.255	To create ACE (Access Control Entry). In this example you specified **ONLY** the permit/deny, network and its wildcard mask. The default sequence number will be 10, 20, 30, ...

remark “PUT YOUR REMARKS HERE”	For notes and documentation on how the ACL work
ip access-group 1 { out | in }	Applying the ACL in the interface router (numbered standard ACL way)
ip access-group FOR-192.168.1.0/24 { out | in }	Applying the ACL in the interface router (named standard ACL way)

What I have learned:

→ How to create standard { numbered | named } access control list (ACLs).
→ How to create access control entries (ACE) for access control list (ACLs).
→ How to apply the access control list (ACLs) to interface

Verification:

→ 172.16.1.0/24 and 172.16.2.0/24 network can’t ping each other because of the ACL
→ **ONLY** PC1 (172.16.1.1/30) and PC3 (172.16.2.1/30) can access server 1
→ 172.16.2.0/24 can’t access server 2 while other networks can
→ The OSPF is working since each end hosts can ping each other gateways